6.1 Polynomials Whose Roots Are Roots of Unity

WORKSHEET

The question

For any given positive integer n, a number a (which may be complex) is said to be a *primitive nth root of unity* if $a^n = 1$ and there is no integer m such that $0 < m < n$ and $a^m = 1$. Write down the two primitive 4th roots of unity.

Let $C_n(x)$ be the polynomial such that the roots of the equation $C_n(x) = 0$ are the primitive nth roots of unity, the coefficient of the highest power of x is one and the equation has no repeated roots. Show that $C_4(x) = x^2 + 1$.

(i) Find $C_1(x)$, $C_2(x)$, $C_3(x)$, $C_5(x)$ and $C_6(x)$, giving your answers as unfactorised polynomials.

(ii) Find the value of n for which $C_n(x) = x^4 + 1$.

(iii) Given that p is prime, find an expression for $C_p(x)$, giving your answer as an unfactorised polynomial.

(iv) Prove that there are no positive integers q, r and s such that $C_q(x) \equiv C_r(x)\,C_s(x)$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Every nth root of unity satisfies $z^n = 1$. But the smallest positive power that gives 1 might be less than n. For each of the four 4th roots of unity, 1, i, -1, $-\mathrm{i}$, what is that smallest power?

STAGE 1 The first few

Ask yourself Which roots of unity have order 1, 2, 3, 5 and 6? (Draw the 6th roots of unity and sort them by order.)

STAGE 4 No cyclotomic polynomial is a product of two others

Pause and try Suppose $C_q = C_r C_s$. Take a root of C_r. What is its order? What does the identity say about its order? What does that force?

Try this next

1. **Check the factorisation.** Verify that $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = x^6 - 1$, and that $x^{12} - 1$ factorises as in Figure 6.1.1 (in the full section online).

2. **Twice a prime.** Show that if p is an odd prime then $C_{2p}(x) = C_p(-x)$. Check this for $p = 3$.

3. **Totients.** Show that $\varphi(p^k) = p^k - p^{k-1}$ for a prime p, and that $C_{p^k}(x) = C_p\left(x^{p^{k-1}}\right)$. Use this to find C_9.

4. **A longer one.** HARDERUsing $x^{15} - 1 = C_1 C_3 C_5 C_{15}$, show that $C_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$.

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). The standard undergraduate introduction; cyclotomic polynomials and the regular polygon appear early.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Includes roots of unity and their arithmetic.
- **Marcus du Sautoy, *Finding Moonshine*** (Fourth Estate). A popular account of symmetry and group theory, including the story of Galois.

The full guided solution, and what the question leads to at university, is in Section 6.1:
https://beyondstep.keithmcnulty.org/ch06/cyclotomic.html